

Minutes Document 23

Date: 07/03/2024

Time: 2:00 PM - 3:30 PM

Location: Djanogly Library, Jubilee Campus, University of Nottingham

Present (in order of attendance):

- **Andreas** - Team Admin
- **Aaron** - Deputy Team Admin
- **Daniel** - Deputy Git Master
- **Jacob** - Deputy Team Leader
- **Mateusz** - Git Admin
- **Martin** - Team Leader
- **James Perrott** - Company contact

Agenda:

- Welcome and introduction (1 minutes):

- Call the meeting to order
- Converse with present team members
- Briefly introduce the purpose of the meeting

- What needs to be discussed (10 minutes):

- The direction of the AI team and how they will be progressing from this point
- The team is going with the carbon emissions predictor/calculator use case
- We will be using MLM to do this, make the calculations and the predictions
- How we will be doing this - Jacob explaining what he plans to do with the AI team

- What was discussed in the meeting (50 minutes):

- In any AI project we ask if AI is the best way of doing this - Machine learning Models?
- Website to look at:
 - WebTAG - transport analysis guidance - guidance given to civil engineers on evaluate transport projects - free to access
 - Post Office for every post codes
 - Can use work calendar for locations
 - Amazon vector database - the vectors will include the start, finish, distance and time.
- The models to use - origins and destinations - vector database - find vectors that are similar to each other - the co-sign distance between them, the angle - are a bit slow - we need to make sure we look at calculating that distance between the vectors, so the same angle and distance. How to get the data locations
 1. Map
 2. Pick random destination (1 city)
 3. Snap to nearest road
 4. Repeat for multiple routes (thousands?)
- Use a vector search to underpin that - not strictly machine learning but using similar techniques.
- One working forwards and working backwards
 - Working backwards - something that will have an impact - the use case we have come up with

- Working forwards - how the platform will affect how the user will travel
- feasibility - what data is available how to use it and what can we actually do - work in some former thinking, so they type of machine learning we are doing
- Can look into making an artificial dataset - start with a map and then pick random locations away from their work and take into account that distance, so the traffic.
- let each employee control their own data - ask them if they want to be matched with other prop and their data to be shared - make sure that the data is hashed, so secure holding of the data.
- Want a Journey table with every individual journey - access to the user table
 - UserTable (ID, hashpass?)
 - JourneyTable, dest, origin, startTime,EndTime
- Can tell the users that their data expires in 2 years for data security
- Look at one city - London

- What he is expecting:

- see if we can have a modular approach - work out what info needs to be passed from one function to another and the self obtain those units
- Make sure the code can be easy to be upgraded and maintainable
- Unit testing
- He can't say for certain as he don't know how we will be marked
- Start simple, making sure everything is semi-independent and then join them at the end - needs to look at specifying these components and then share out the jobs

- What we need to do for Tuesday (45 Minutes):

- A full use case and what the app will be doing specifically - Martin
 - Work on data visualisation and see how it should look
- Split up these components and then share out the jobs - Everyone
- Make a big outline document - Andreas - make sure it is on track
- **AI Team** - Research and implementation of Vector databases - one or two origin points - have documentation for this research and implementation
- **UI Team**
 - Change the dashboard so that the user can insert their data there instead of having the separate data manager page.
 - Form on the left with the MLM button below and then the results on the right hand side - the form should have menus to select
 - Trying to find a way to access the user's location from the webapp
 - See if we can send the results on the email once we get the results on the webapp.
 - Broader question - if James can help us with any industry contracts - ask him on contact

Next Meeting: Tuesday 12th March - Time: 5:00PM - Place: Computer Science Building

Submitted by Andreas Sergiou